Practical No. 8

Aim: To implement a typical service and a typical client using WCF.

Part A — Create the WCF Service

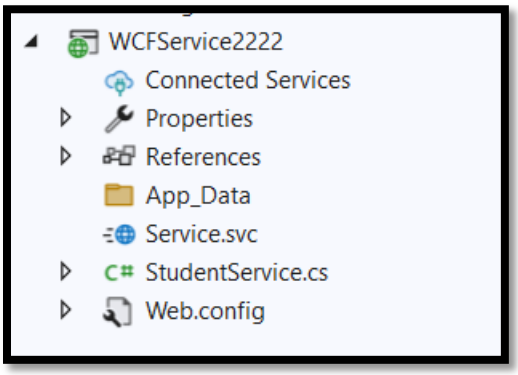
Create:

WCF Service Application (.NET Framework)

Use **.NET Framework 4.8**.

Suppose the project is named:

WCFService2



Step 1 — Create the service code

Open:

App_Code → StudentService.cs

Replace its contents with:

```
using System;
using System.ServiceModel;

namespace WCFService2
{
    [ServiceContract]
    public interface IStudentService
    {
```

```
    [OperationContract]
    string GetStudentResult(string name, int marks);
  }

  public class StudentService : IStudentService
  {
    public string GetStudentResult(string name, int marks)
    {
      string result;

      if (marks >= 75)
        result = "Distinction";
      else if (marks >= 60)
        result = "First Class";
      else if (marks >= 50)
        result = "Second Class";
      else if (marks >= 40)
        result = "Pass";
      else
        result = "Fail";

      return "Student Name: " + name +
          "\nMarks: " + marks +
          "\nResult: " + result;
    }
  }
}
```

Step 2 — Configure Service1.svc

First remove already added Service1.svc file and then

Right click on your project --- add ----- new items ----- select text file ----- give it name as Service1.svc

Open:

Service1.svc

```
<%@ ServiceHost Language="C#" Debug="true"
    Service="WCFService2.StudentService" %>
```

Step 3 — Configure Web.config

Use:

```
<?xml version="1.0"?>


<configuration>


  <system.web>
    <compilation debug="true" targetFramework="4.8"/>
  </system.web>


  <system.serviceModel>


    <behaviors>
      <serviceBehaviors>
        <behavior name="StudentServiceBehavior">
          <serviceMetadata httpGetEnabled="true"/>
        </behavior>
      </serviceBehaviors>
```

```
    </behaviors>

    <services>
      <service name="WCFService2.StudentService"
               behaviorConfiguration="StudentServiceBehavior">

        <endpoint
          address=""
          binding="basicHttpBinding"
          contract="WCFService2.IStudentService" />

      </service>
    </services>

    <serviceHostingEnvironment
        aspNetCompatibilityEnabled="true"
        multipleSiteBindingsEnabled="true" />

  </system.serviceModel>

</configuration>
```

Step 4 — Build the service

Go to:

Build → Build Solution

or press:

Ctrl + Shift + B

You should get:

Build succeeded.

Make sure there are **no errors**.

Step 5 — Run the service

Right-click:

Service1.svc

Select:

Set as Start Page

Then press:

Ctrl + F5

The browser should open something similar to:

http://localhost:xxxxx/Service1.svc

You should see a WCF page saying approximately:

You have created a service.

You should also see a **WSDL** link.

Test:

http://localhost:xxxxx/Service1.svc?wsdl

If you see XML/WSDL information, your service is working.

Part B — Create the Client

Step 6 — Add Console Client

Right-click your solution:

Solution 'WCFService2'

Select:

Add → New Project

Choose:

Console App (.NET Framework)

Name it:

WCFClient

Step 7 — Add Service Reference

Right-click:

WCFClient

Select:

Add → Service Reference

Enter the service URL:

http://localhost:xxxxx/Service1.svc

Use the **actual port number** generated on your computer.

Click:

Go

Visual Studio should discover:

StudentService

GetStudentResult

Set:

Namespace:

StudentServiceReference

Click:

OK

Step 8 — Client Program

```
using System;

namespace WCFClient
{
    class Program
    {
        static void Main(string[] args)
        {
            Console.Write("Enter Student Name: ");
            string name = Console.ReadLine();

            Console.Write("Enter Marks: ");
            int marks = Convert.ToInt32(Console.ReadLine());

            StudentServiceReference.StudentService client =
                new StudentServiceReference.StudentService();

            string result = client.GetStudentResult(name, marks);

            Console.WriteLine();
            Console.WriteLine("----- STUDENT RESULT -----");
            Console.WriteLine(result);

            client.Close();
```

```
            Console.ReadKey();
        }
    }
}
```

Step 9 — Run Both Projects

Right-click the solution:

Properties → Common Properties → Startup Project

Select:

Multiple startup projects

Set:

Project	Action
WCFService2	Start
WCFClient	Start

Click:

Apply → OK

Then:

Ctrl + F5

